

$$\bar{A}(M) = \begin{pmatrix} 1 & -3 \\ 0 & -3 \end{pmatrix} \cdot M + M \cdot \begin{pmatrix} -1 & 0 \\ 1 & 3 \end{pmatrix}$$

$M_{2 \times 2}$

$A = ? \quad \text{Ker } \bar{A} = ?$

$\dim \text{Ker } \bar{A} = ?$

$$y_1 = 0 \cdot x_1 - a_3 x_2 + a_2 x_3$$

$$y_2 = a_3 x_1 - 0 \cdot x_2 - a_1 x_3$$

$$y_3 = -a_2 x_1 - a_1 x_2 + 0 \cdot x_3$$

$$A = \begin{pmatrix} 0 & -a_3 \\ a_3 & 0 \\ -a_2 & a_1 \end{pmatrix}$$

$$(-1)^3 \cdot (-a_3)$$